# Machine Learning for Spectral Analysis

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## 1. Introduction

Optical Emission Spectroscopy (OES) provides **non-invasive, real-time diagnostics** of plasma processes by analysing emission spectra from excited species. Plasma-based manufacturing (semiconductor etching, surface treatment, thin-film deposition) relies on OES for process monitoring, but manual spectral interpretation is slow and error-prone.

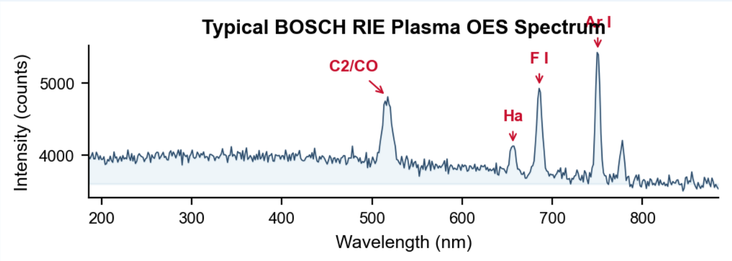
This project develops an **automated ML pipeline** for plasma OES analysis targeting four objectives:

1. **Spectral feature identification** — automated peak detection, NMF decomposition, NIST database matching
2. **Plasma species classification** — SVM, RF, CNN, Transformer comparison (6 models)
3. **Spatiotemporal evolution** — Attention-LSTM phase prediction, species time-series extraction
4. **Semi-quantitative intensity** — actinometry (Coburn & Chen 1980), Boltzmann Te estimation

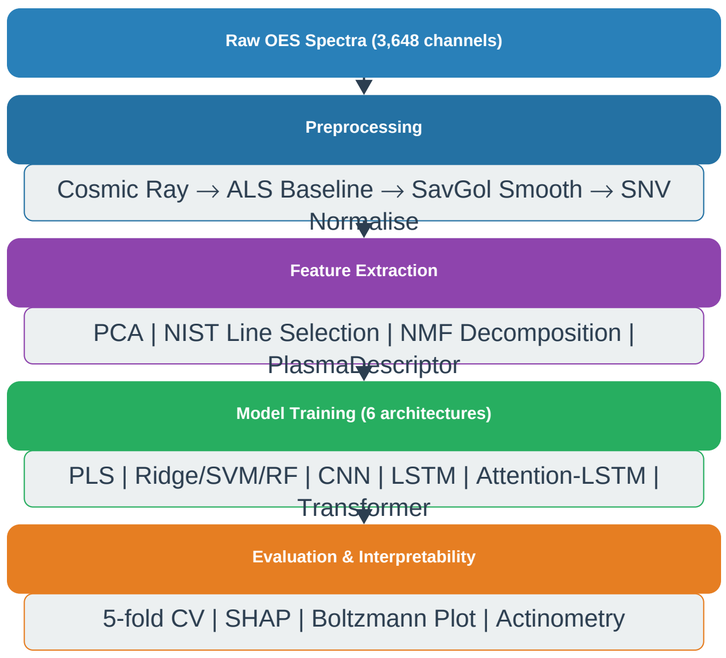
**Industrial relevance:** Enables real-time closed-loop control of plasma processes in semiconductor fabrication, reducing defect rates and improving yield.

### Three public datasets:

| Dataset | Channels | Notes |
|---|---|---|
| LIBS Benchmark | 40,002 ch | Steel composition (12 classes) |
| Mesbah CAP | 51 ch | N2 plasma T_rot / T_vib |
| BOSCH RIE | 3,648 ch | 25 Hz, 10 days, SF6/C4F8 |




## 2. Methodology




*Hyperparameter optimisation: Optuna two-stage search (20 trials per target)*

### Key Design Decisions

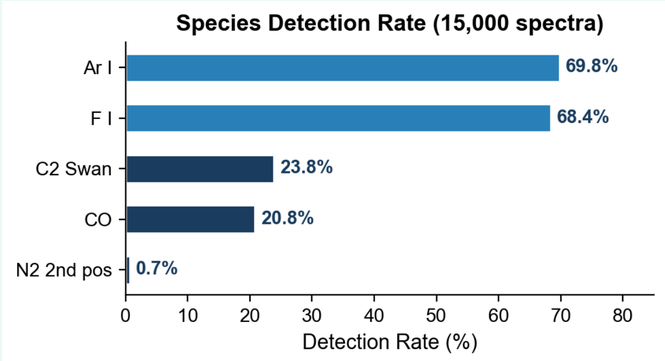
- **Per-element model routing:** Different plasma species require different model architectures. Cr uses Ridge+PCA (avoids overfitting on 976 NIST lines); other elements use ANN+NIST windows.
- **GroupKFold CV:** Prevents same-target spectra leaking between train/test folds. Forces model to generalise across physical samples, not memorise measurement noise.
- **Balanced class weights:** Plasma OFF samples are only 12.1% of data. Class-weighted loss prevents majority-class bias in all classifiers.
- **NMF over PCA:** Non-negative components are physically interpretable as pure-species emission spectra. PCA components can be negative, violating emission physics.

**Preprocessing rationale:** ALS baseline ($\lambda=10^5$) removes fluorescence continuum. Savitzky-Golay (window=11, order=3) preserves peak shapes while reducing shot noise. SNV normalisation corrects for

## 3. Species Identification

**Non-negative Matrix Factorization (NMF)** decomposes the spectral matrix **X** ≈ **W · H**, where each row of **H** is a pure-species emission spectrum and **W** contains the corresponding concentrations. NMF is physically appropriate because emission intensities are inherently non-negative.

**Automated NIST matching:** Each detected peak is compared against 39 reference emission lines from 13 species. The algorithm selects the closest database match within +/-1.5 nm tolerance. Species with peak intensity > $\mu + 3\sigma$ (global spectrum statistics) are classified as *present*.

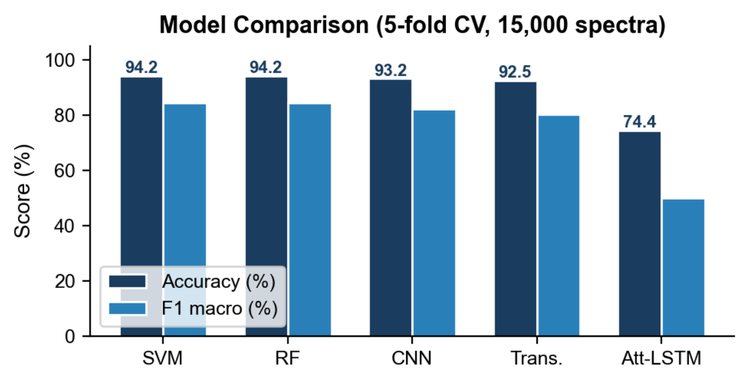


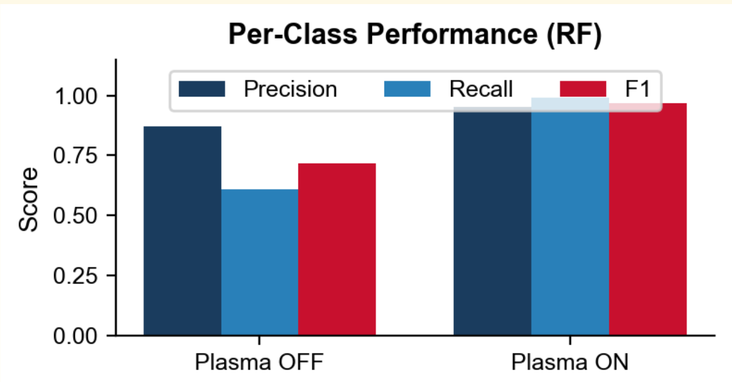

| Species | Lines (nm) | Origin |
|---|---|---|
| F I | 685.6, 690.2, 703.7 | SF6 etchant radical |
| Ar I | 696.5, 750.4, 763.5 | Carrier gas reference |
| C2 Swan | 473.7, 516.5, 563.6 | C4F8 carbon indicator |
| CO | 451.1, 519.8, 561.0 | C4F8 + O2 product |
| Si I | 250.7, 252.4, 288.2 | Si etch product |
| SiF | 440.0, 442.5 | Si + F recombination |

**Validation:** NMF Component 0 peaks at 684.4 nm (≈ F I 685.6); Component 2 at 515.1 nm (approx. C2 Swan 516.5) — unsupervised decomposition confirms NIST species independently.

## 4. Classification Results

**94.2% Accuracy (SVM/RF, 5-fold CV)**



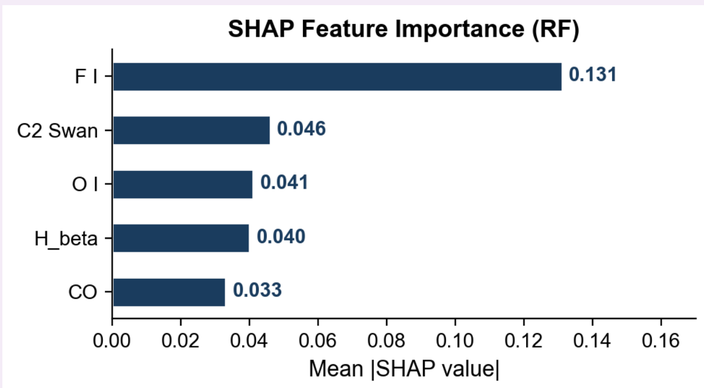

### Species Detection (13 species):

| Species | Rate | Species | Rate |
|---|---|---|---|
| Ar I | 69.8% | C2 Swan | 23.8% |
| F I | 68.4% | CO | 20.8% |

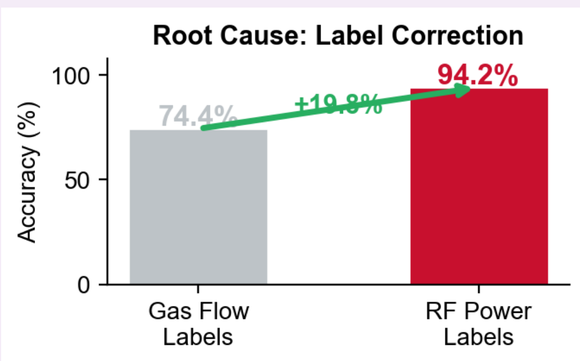
### Model Architectures:

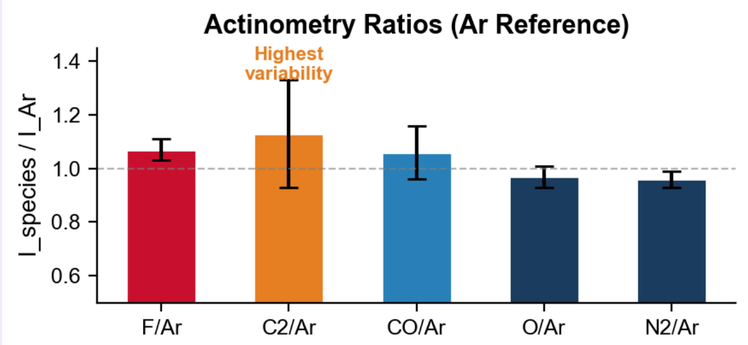
- **SVM/RF:** StandardScaler + RBF kernel (C=10) / 200 trees. Balanced weights.
- **CNN:** Conv1D(32-64-128) + AdaptiveAvgPool + FC(64) + Dropout(0.3).
- **Transformer:** 1D patch(64), d=128, 4 heads, 3 layers. [CLS] + AdamW.
- **Att-LSTM:** 2-layer LSTM(64) + additive attention. PCA(20) windows.

## 5. Interpretability & Physics




F I (fluorine radical) = most discriminative species, consistent with SF6 etchant chemistry.




**Boltzmann T_exc = 13,334 K** (6 Ar I lines). **DTW K-Means (k=4):** 4 discharge phases identified, F I 684 nm emission ratio 2.04x between clusters.

## 6. Conclusions & Further Work

### Key Achievements:

✓ Automated species detection: 13 plasma species, 39 NIST emission lines, NMF decomposition validates against database

✓ 94.2% plasma state classification (SVM/RF best; CNN 93.2%, Transformer 92.5% — 6 models compared)

✓ SHAP interpretability: F I importance = 0.131 (3x next species), physically validated as primary SF6 etchant radical

✓ Data-driven label correction: gas-flow labels (74%) → RF-power labels (94%) via root-cause spectral analysis

✓ Temperature regression: T_rot RMSE = 20.0 K, T_vib = 102.0 K (Mesbah CAP dataset)

✓ 78 automated tests, 6 CLI task modes, fully reproducible pipeline

### Limitations:

- OES→process parameter regression failed (R2 < 0, causal irreversibility)
- Boltzmann Te limited by narrow Ar I energy spread (0.33 eV)
- Spatial etch prediction lacks wafer ID alignment

### Further Work:

- Extend to multi-class species classification (beyond binary ON/OFF)
- Implement real-time OES monitoring with streaming inference
- Apply transfer learning across different plasma reactors

**Project Metrics:** 15,000 spectra analysed | 3,648 spectral channels | 13 species identified | 39 NIST emission lines | 6 ML models compared | 78 automated tests | 32 development stories | 23 literature references | 6 CLI task modes | 3 public datasets

### References:

[1] Gidon *et al.* (2019) IEEE Trans. Radiat. Plasma Med. Sci.

[2] Coburn & Chen (1980) J. Appl. Phys. — Actinometry

[3] Vaswani *et al.* (2017) NeurIPS — Transformer

[4] Contreras *et al.* (2024) Anal. Chem. — Spectral zone